

Name: Connor Harris, Bennett Lohmeyer, Nic Blucher

Date: __12/10/24__ **Class:** 8

Element B

Prior Design Attempt #1 - <Wagon Handle Extension>

Source(s):

<https://www.aliexpress.us/item/3256805866898601.html?src=google&gatewayAdapt=glo2usa>

<https://www.google.com/>

<https://patentimages.storage.googleapis.com/57/94/93/a423b65678d722/US10988153.pdf>



Details:

The handle is fairly universal; however, the main use for this design incorporates pulling to take the pressure off of your lower lumbar. The idea is that the wheels would swivel at the

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front allowing for the handle to serve as an attachment to the human for the cart to follow. This handle can be attached to trolleys to make it easier for people to pull the trolley. There are different lengths that the handle can be set at so people of different heights and arm lengths can pull comfortably and with ease. The Wagon Cart Pull Handle Replacement, with its adjustable angle and telescoping rod, is designed to enhance maneuverability and convenience for outdoor trolleys, making it an ideal solution for camping needs (CampingSupplies, 2024)."

Analysis

Though this is a successful design for pulling a trolley, our goal is to make a handle for a cart that makes it more comfortable to push. This solution solves a slightly different problem than the one that we are attempting to fix since the stakeholder is trying to push our cart, not pull it.

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Prior Design Attempt #2 - <Shopping Cart with Dual Swivel Wheels for Groceries with Adjustable Handle Height>

<https://thegivenget.com/products/shopping-cart-with-dual-swivel-wheels-for-groceries-compact-folding-portable-cart-saves-space-with-adjustable-handle-height-lightweight-easy-to-move-holds-up-to-70lb-max-66lbs>

<https://patents.google.com/patent/US5899285A/en>



Details

This handle is securely fastened to any cart. The idea behind this design is that it is adjustable to certain heights so that people of all heights can reach the handle and handle it with some comfort. On the other hand, the only adjustable part of the design is the

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ladder and not the angle at which it pushes. This compact folding shopping cart stands out for its dual swivel wheels, which provide superior maneuverability in confined spaces such as grocery store aisles. Its lightweight design, combined with the adjustable handle height, ensures user convenience and comfort. Supporting up to 70 liters or 66 pounds, the cart offers significant capacity without compromising on ease of movement. The folding mechanism adds a layer of practicality, allowing users to store the cart efficiently when not in use, making it ideal for urban living or limited storage spaces(ShoppingCart Co., 2024).

Analysis

This type of cart handle is useful for people of most heights. The only problem is that there is limited adjustability. With only three different settings, some people who are really tall or short may be excluded from the range of this handle. This poses an issue for the people in between two heights. Another issue with this design is that it doesn't satisfy the desire to be easily removable. This design is faceted in place and doesn't allow one to remove without the use of tools provided by the provider upon purchase.

Prior Design Attempt #3 - <Utility Cart Handle>

Shelving Inc.

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<https://patentimages.storage.googleapis.com/ef/80/bb/ea22bac59370e9/USD948833.pdf>



Manufacturer: Shelving Inc.

Details

This adjustable cart handle makes it easy for taller people to push a cart around without having to bend over too much. The handle stands a tall 35 inches, allowing for the optimal height of an average person to use the car with comfort. The Utility Cart Handle is a practical and innovative product designed to transform stationary wire shelving into a mobile utility cart, offering enhanced versatility and functionality. The handle stands at 35 inches tall, providing a comfortable height for maneuverability, and features grooves

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spaced in 1-inch increments, allowing users to customize shelf placement to accommodate items of various sizes (UtilityCart Co., 2024).

Analysis

This handle doesn't adjust to different heights, the fact that there is only one height limits usability for different persons. People who are too tall or too short still may struggle to use this handle comfortably. The goal of our product is to make it usable for pretty much all heights. While this design stands at climbing heights, it only reaches about 3 feet tall which is extremely problematic in a situation where the client exceeds 6'.

Prior Design Attempt #4 - <Cart Push Handle Ergonomic Replacement Trolley Handle Wagon Trolley Cart Attachment>

JECOMPRIS; Vender: Amazon

<https://www.amazon.com/JECOMPRIS-Ergonomic-Replacement-Trolley-Attachment/dp/B0D7HWL>

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Details

The ergonomic replacement trolley attachment from JECOMPRIS offers a highly functional design that prioritizes user comfort and efficiency. The attachment's adjustable components enhance load distribution, making it ideal for heavy-duty applications in logistics and personal use. Its focus on durability and adaptability allows users to replace damaged or inefficient trolley parts with minimal effort. By addressing user ergonomics, this product caters to a diverse range of activities, from warehouse work to outdoor tasks, significantly improving productivity and ease of use."The ergonomic replacement trolley

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attachment from JECOMPRIS improves load management and user comfort through its innovative design, making it suitable for heavy-duty applications (JECOMPRIS, 2024)."

Analysis

Though this is a successful design for pulling a trolley, our goal is to make a handle for a cart that makes it more comfortable to push. This solution solves a slightly different problem than the one that we are attempting to fix. Although the handle is adjustable and removable, the cons of the product present bigger issues for the higher-ranked criteria.

Prior Design Attempt #5 - <insert name of prior design attempt #5>

<https://www.walmart.com/ip/2pcs-Trolley-Cart-Handle-Replacement-Trolley-Grip-Handle-Shopping-Cart-Part-Handle/10307518308?wmlspartner=wlpa&selectedSellerId=102548276>



Details

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This set of trolley grip handles serve as a cost-effective solution for restoring the usability of worn-out or damaged shopping carts. The handles are designed for easy installation, offering a secure and comfortable grip for users. The durable construction ensures long-term reliability, while the universal design allows compatibility with a wide range of cart models. These replacement handles not only enhance the functionality of existing carts but also extend their service life, reducing waste and promoting sustainability (TrolleyParts Co., 2024).

Analysis

This product is extremely effective; however, the utilization of this product is not necessarily effective for pushing/pulling carts of this size. This product focuses primarily on moving lightweight carts or luggage bags. The idea is ergonomic, but similar to the other designs fails to accomplish some of the higher-ranked criteria tasks.